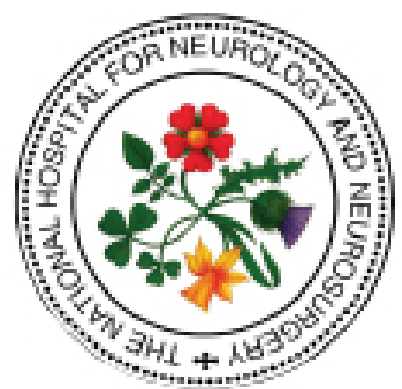
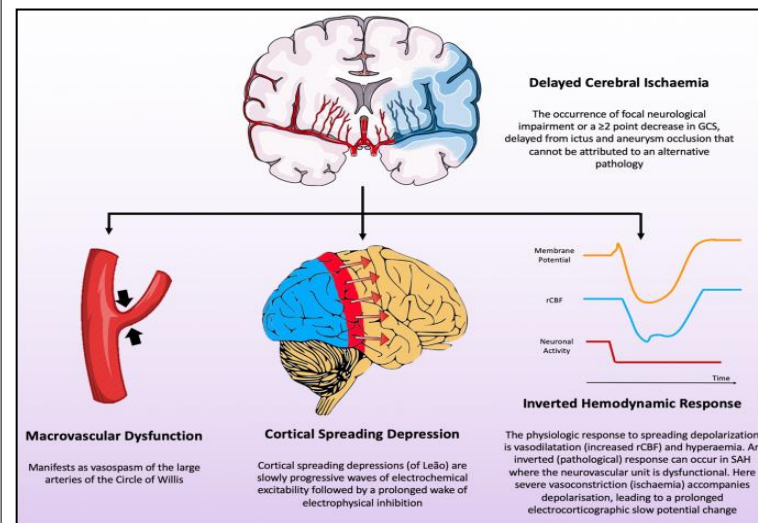




MONITORING / DETECTION / MANAGEMENT of CEREBRAL VASOSPASM and DCI in treated ANEURYSMAL SUBARACHNOID HAEMORRHAGE (aSAH)

Class of Recommendation

- ▶ **Class 1: STRONG**
Recommended (Benefit >>> Risk)
- ▶ **Class 2a: MODERATE**
Reasonable (Benefit >> Risk)
- ▶ **Class 2b: WEAK**
May be reasonable (Benefit ≥ Risk)
- ▶ **Class 3: STRONG**
Potential harm (Risk > Benefit)



CEREBRAL VASOSPASM (narrowing of cerebral arteries - risk period usually beginning 4-10 days after aSAH and lasting for several days) if prolonged, can lead to **DELAYED CEREBRAL ISCHAEMIA (DCI)** and infarction and is most common cause of late morbidity and mortality.

Clinical deterioration due to DCI is defined as focal neurological impairment or decrease of at least 2 GCS points lasting for >1h, not apparent immediately following aneurysm treatment, and not attributable to other causes.

CURRENT VASOSPASM GUIDELINES

<https://www.ahajournals.org/doi/suppl/10.1161/STR.0000000000000436>

Prompt investigation of source of haemorrhage and treatment of any ruptured aneurysm recommended within 24h of bleed to prevent rebleeding and manage the complication of vasospasm.

BP AUGMENTATION

Although for many years induced hypertension (iHTN) has been key in management of DCI, lack of quality evidence and potential complications of hypertensive therapy have led to renewed debate.

Current guidelines state that in symptomatic vasospasm it may be reasonable to elevate BP to reduce progression and severity of DCI but do not propose a specific target. Any manipulation of BP should be carefully considered in light of individual haemodynamics and be limited in extent and duration.

FLUID THERAPY – Aim for EUVOLAEMIA

Maintaining euvolaemia in symptomatic vasospasm can be beneficial in reducing progression and severity of DCI. However, **prophylactic** haemodynamic augmentation and hypervolaemia are not recommended.

MULTIPLE ANEURYSMS

Any additional unsecured aneurysm **not** thought to be responsible for the acute SAH, should not influence haemodynamic management of vasospasm.

MONITORING / DETECTION / MANAGEMENT

CLINICAL ASSESSMENT

- GCS, Pupils, Limbs – additional monitoring needed when neuro exam not possible, as in high grade aSAH
- **SAVE** signs

MONITORING FOR CEREBRAL VASOSPASM AFTER ANEURYSMAL SUBARACHNOID HAEMORRHAGE (SAH)
'SAVE' signs or symptoms may be present without a change in GCS

S	Severe persisting headache resistant to analgesia Subtle changes in speech – Is patient able to name objects?
A	Arm drift – Is patient able to equally maintain both arms outstretched, palms upwards with eyes closed?
V	Visual acuity – Is patient able to read as well as prior to SAH?
E	Elevated BP – Is BP elevated? (Sudden rise in BP often premonitory sign of vasospasm)
If any SAVE sign or symptom present inform nurse in charge / medical team immediately & increase frequency of observations. Also escalate if patient has a change in behavior, such as VAGUENESS, AGITATION or CONFUSION. Monitor more closely if patient's relatives report any change	

DIAGNOSTIC TOOLS

- Daily Transcranial Dopplers (TCDs)
- CTA / CTP / PbtO₂

Since vasospasm is diagnosis of exclusion, if new neurological deficit occurs CTH needed to exclude hydrocephalus or infarct.

NIMODIPINE 60mg / 4 hourly NG or PO

- Commence as soon as SAH diagnosed
 - Continue for 21 days
- Early enteral nimodipine beneficial in preventing DCI & improving outcome. Consistent administration recommended even in cases of nimodipine-induced hypotension
- If hypotensive following dose change to 30mg / 2 hourly
 - If not absorbing, give IV (1-2mg/h) via dedicated CVC lumen (to run concurrently with 0.9% saline @ 40mL/h)

FLUID THERAPY – Targeted fluid management

- Aim EUVOLAEMIA with IV crystalloid to help prevent DCI & improve outcome
- 2-3L / 24h for most patients - consider individual needs
- Institute cardiac output monitoring (FloTrac) with PPV to guide volume resuscitation where appropriate
- If IV bolus needed give 250mL (100mL if cardiac history)

BP MANAGEMENT

- SBP target range ≈120-160mmHg
- If patient self-hyper tensing do not attempt to lower BP
- Permissive hypertension reasonable given that patients developing DCI often show rise in BP that correlates with onset of cerebral vasospasm

Does patient have a **CHANGE** in **NEUROLOGY** or any **SAVE** sign suggesting **VASOSPASM**?

OTHER
CAUSES
EXCLUDED
BY CTH

NO

YES

**CONTINUE
MONITORING /
PREVENTIVE
MEASURES**

STEP 1 INDUCED HYPERTENSION (iHTN) (secured aneurysm)

In symptomatic vasospasm, elevating SBP may be reasonable to reduce progression and severity of DCI

1. Stepwise augmentation of SBP to reverse neurological deficit - **20mmHg increments above baseline each hour up to 180mmHg**
2. Review after **6h**
3. if improvement, set **new BP target:** based on response to augmenting BP
4. If no improvement, employ adjuvant therapy - NCC Consultant / Neurosurgeon discuss need for imaging with Interventional Neuroradiologist (INR)

1. **Metaraminol** (see protocol) - If required >24h, insert CVC with **purple port** & switch to noradrenaline
2. **Noradrenaline** (see protocol) - Make stepwise BP increases (in 20mmHg increments above baseline) & titrate to neurology to determine BP target

DURATION of iHTN

- Avoid excessive & prolonged BP augmentation
- Daily review of hypertensive therapy & TCD results
- Consider trial of lowering BP target to determine continued need for iHTN
- Stepwise vasopressor reduction based on neurology

▶ Patients 'at risk' of DCI should **not** have **prophylactic** BP augmentation - to reduce risk of iatrogenic patient harm

STEP 2 +/- INTERVENTIONAL NEURORADIOLOGY

If no reversal of deficit after 6h of iHTN, NCC Consultant / INR / Neurosurgeon discuss imaging options

1. CTA / CTP
2. Angiography

STEP 3 +/- ENDOVASCULAR OPTIONS

1. Angioplasty
2. Direct intra-arterial injection with vasodilators

STEP 4 +/- MEDICAL TREATMENT

1. Brain Tissue Oxygen (PbtO₂ bolt – See protocol)
2. Milrinone (See protocol)